



CEN111

INTRODUCTION

Dr. Barış ATA



CEN111 Algorithms and Programming I

- Fall 2021-2022
- Instructor: Dr. Barış ATA (bata@cu.edu.tr)
- Office Hours: Wednesday 13:00-15:00
- Grading: Midterm 40 %, Final exam 60 %.

Description

- This course aims to teach students the basics of computer programming using C Programming Language.
- This course focuses on
 - *Algorithm design tools: flowcharts, pseudocodes*
 - *C programming language primitives*
 - *I/O functions*
 - *Operations*
 - *Conditions*
 - *Loops*
 - *Functions*
 - *Arrays*

Textbooks

- Problem Solving and Program Design in C —7th ed,
 - *Jeri R. Hanly, Elliot B. Koffman*
- C How to Program, 8th ed,
 - *Paul Deitel, Harvey Deitel*

References

- C Programming Language -2nd Edition
 - *Brian W. Kernighan, Dennis M. Ritchie*
- C Programlama Dili
 - *Rifat Çölkesen*
- C Dersi: Programlamaya Giriş
 - *Nergiz Ercil Çağıltay, C. Fügen Elbes, Gül Tokdemir, Çiğdem Turhan*

Required Software

- Raptor Flowchart based Programming Environment
- Microsoft Visual Studio

Computers

- Hardware
 - *the actual computer equipment*
- Software
 - *the set of programs associated with a computer*

Computers

- Program
 - *a list of instructions that enables a computer to perform a specific task*
- Binary number
 - *a number whose digits are 0 and 1*

Computer Hardware

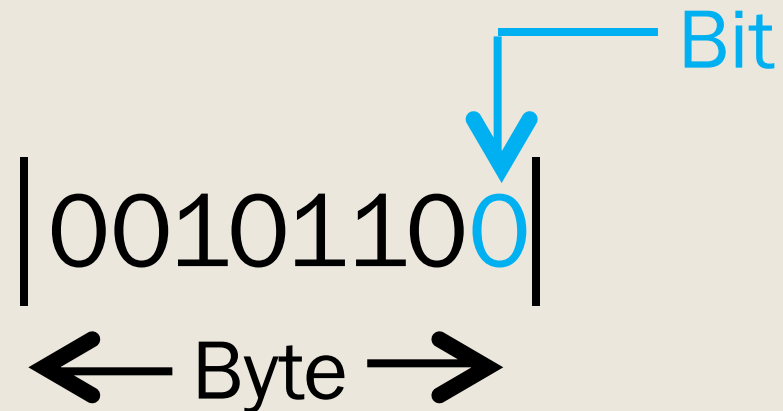
- Main Memory
- Secondary Memory
- Central Processing Unit
- Input Devices
- Output Devices

Memory

- Memory cell
 - *an individual storage location in memory*
- Address of a memory cell
 - *the relative position of a memory cell in the computer's main memory*
- Contents of a memory cell
 - *the information stored in a memory cell, either a program instruction or data*

Memory

- Stored program concept
 - *a computer's ability to store program instructions in main memory for execution*
- Byte
 - *the amount of storage required to store a single character*
- Bit
 - *a binary digit, a 0 or a 1*



Memory

- Random access memory (RAM)
 - *the part of main memory that temporarily stores programs, data, and results*
- Read-only memory (ROM)
 - *the part of main memory that permanently stores programs or data*

Secondary Memory

- Secondary storage
 - *units such as disks or flash drives that retain data even when the power to the drive is off*
- Disk
 - *thin platter of metal or plastic on which data are represented by magnetized spots arranged in tracks*
- Optical drive
 - *device that uses a laser to access or store data on a CD or DVD or Blu-ray Disk*
- Flash drive
 - *device that plugs into USB port and stores data bits as trapped electrons*

Central Processing Unit

- Central Processing Unit (CPU)
 - *coordinates all computer operations and performs arithmetic and logical operations on data*
- Fetching an instruction
 - *retrieving an instruction from main memory*
- Register
 - *high-speed memory location inside the CPU*
- Multiprocessor
 - *a computer with more than one CPU*

Input/Output Devices

- Cursor
 - *a moving place marker that appears on the screen*
- Function keys
 - *special keyboard keys used to select a particular operation; operation selected depends on program being used*
- Mouse, touchpad
 - *input devices that move a cursor on the computer screen to select an operation*

Computer Software

- Operating System
- Application Software
- Computer Languages
- Executing a Program

Operating System

- Operating system (OS)
 - *software that controls interaction of user and computer hardware and that manages allocation of computer resources*
- Booting a computer
 - *loading the operating system from disk into memory*

Application Software

- Application

- *software used for a specific task such as word processing, accounting, database management, playing a game, or checking the weather forecast*

- Install

- *make an application available on a computer by copying it to the computer's hard drive*

Computer Languages

- Machine language
 - *binary number codes understood by a specific CPU*
- Assembly language
 - *mnemonic codes that correspond to machine language instructions*
- High-level language
 - *machine-independent programming language that combines algebraic expressions and English symbols*

Computer Languages

- Compiler
 - *software that translates a high-level language program into machine language*
- Source file
 - *file containing a program written in a high-level language; the input for a compiler*
- Syntax
 - *grammar rules of a programming language*
- Object file
 - *file of machine language instructions that is the output of a compiler*

Computer Languages

- Linker
 - *software that combines object files and resolves cross-references to create an executable machine language program*
- Integrated development environment (IDE)
 - *software package combining a word processor, compiler, linker, loader, and tools for finding errors*

The Software Development Method

1. Specify the problem requirements.
2. Analyze the problem.
3. Design the algorithm to solve the problem.
4. Implement the algorithm
5. Test and verify the completed program.
6. Maintain and update the program.